

Modbus Lab Incident Report

What Happened and When?

When did the incident occur?	06/16/2025, 09:53:56 - 09:59:16 AM
What was the nature of the incident?	Malicious commands sent to PLC.
Which specific assets were at risk?	HMI (A0002), Control Server (A0007), PLC (A0003) and corresponding field devices
What is the severity or significance of the incident?	Threat actor could cause physical damage to plant and/or faculty.

What was the Root Cause?

What caused the incident?	Attacker likely exploited a vulnerability in HMI.
How do we know?	Analysis of Suricata logs show attacker was able to send unauthorized messages to PLC (T1692.001).

What Was and Remains to Be Done?

Identification: How was the problem detected?	Attack was detected via Suricata IDS.
Containment: How will we be able to limit the incident's scope, including adverse effects on the affected data and systems?	Isolate the HMI and PLC.
Eradication: What steps need to be taken to eliminate adversarial presence from the affected environment, protect the affected data, or minimize the risks to the affected parties?	Reset all passwords in the environment and verify all ACL and firewall configurations. If necessary (e.g. rootkit), rebuild with new hardware.
Recovery: How can we restore normal business operations or normal activities?	Ensure safety and control systems are operating as expected. When ready, bring interdependent systems back up properly.